DevSync

رواد مصر الرقمية

GRADUATION PROJECT

Project Planning & Management.

DevSync: Secure AI Project Collaborator

Flutter graduation project | Planning phase: 2 Oct - 16 Oct 2026

TEAM OF 5

Donia Alhosin

Bishay Talaat

Yousef Mohy

Emad Hany

Saga Maher

REPORT DURATION

2 Oct 2026 - 16 Oct 2026

2 weeks

Overview of this version

This version replaces the rotating-slice model with the **Feature Papers**: one owner per feature across UI, logic, Firestore and tests, point loads (47 core points), a backup chain and fixed reviewers. Dates, freezes and scope are unchanged.

Required component	Where to find it
Project Proposal: overview, objectives, scope	Part 1
Project Plan: timeline, milestones, deliverables, resources	Part 2
Task Assignment and Roles	Part 3
Risk Assessment and Mitigation Plan	Part 4
KPIs and course requirements	Part 5
Future work	Part 6

Part 1: Project proposal

DevSync is a Flutter app where developers find open-source collaborators. AI scores each user against projects using their GitHub activity and skills, and prompts are evaluated and tuned with like/skip feedback.

Item	Detail
Project	DevSync: Secure AI Project Collaborator, Flutter graduation project, team of 5
Timeline	Implementation 16 Oct to 30 Nov 2026: six 1-week sprints plus a 4-day final window
Backend	Firebase (Auth, Firestore, Cloud Messaging) and Cloud Functions for notification triggers and AI calls
Platform	Android, Free and Pro flavors
Distribution	Signed, obfuscated build shared by install link (GitHub Releases or Firebase App Distribution), plus an Android App Bundle for the course
Tracking	Jira Scrum board; Git and GitHub with pull requests

Objectives. Working AI matching; secure Google and email sign-in (Firebase UI Auth); FCM alerts; smooth animations; Free and Pro obfuscated builds; every member builds every layer of their own features; full documentation, README and a security-reviewed app.

Must have (protected)	If time allows (after the 12 Nov core freeze)
Sign-in, profile with GitHub data and skills, projects (seed and user-added), AI match feed with score and reason, like/skip, collaboration requests, FCM alerts, core animations, Free and Pro flavors, obfuscated builds and install link, documentation	Task board for accepted collaborations (Donia), task alerts (Yousef), prompt-quality dashboard (Emad), extra animations and usability walkthrough (Bishay), regression tests and Pro-only extras (Saga)

Out of scope: iOS release, store publishing, in-app payments, hosting project code. Technical approach is unchanged: Flutter, Firebase with Cloud Functions (the LLM key stays a function secret), implicit and TweenAnimationBuilder animations, Jira and GitHub.

Part 2: Project plan

Core freeze is 12 Nov and feature freeze is 26 Nov. Milestones are unchanged; the work behind each one is now assigned by feature.

Period	Milestone (end of period)	Features landing
W1: 16-22 Oct	Foundation ready: repo, Flutter app with flavors and theme, Firebase and data model, CI	Foundation (Donia), Design system and wireframes (Donia, done), App shell + CI (Yousef), AI skeleton (Emad), GitHub fetch (Saga)
W2: 23-29 Oct	Sign-in, profile, projects, first AI-ranked feed, like/skip	Auth (Donia), Projects (Bishay), Like/skip and request start (Yousef), AI v1 + feed (Emad), Profile (Saga)
W3-W4: 30 Oct-12 Nov	Requests and FCM end to end (5 Nov), animations in (5 Nov), core complete, security review, first obfuscated build, prompts tuned. Core freeze 12 Nov	Requests + FCM (Yousef), Animations (Bishay), Security rules (Donia), Prompt evaluation (Emad), Build + report (Saga)

Period	Milestone (end of period)	Features landing
W5-W6: 13-26 Nov	If-time items, polish, full testing; buffer if the core is behind. Feature freeze 26 Nov	Task board (Donia), Task alerts (Yousef), Prompt dashboard (Emad), Extra animations (Bishay), Regression + Pro extras (Saga)
Final: 27-30 Nov	Final Free and Pro builds, install link, documentation, README, presentation	Release and README (Saga), Presentation (Bishay), docs by each owner

Deliverables

Deliverable	Lead	Due
Project planning report, brand guide	Donia, Bishay	Done
Core app: all must-have features, tested	All	12 Nov
Test and security report	Saga	12 Nov (final update 30 Nov)
Task board (if time)	Donia	26 Nov
Source code with branches and pull requests	All	Continuous, final 30 Nov
Obfuscated builds, App Bundle, install link	Saga (Yousef supports)	First build 12 Nov, final 30 Nov
Technical documentation and README	Saga assembles; each owner writes their own features	30 Nov
Final presentation and demo	Bishay	30 Nov

Resources. About 12 hours per person per week (roughly 400 team hours). Free tools: Flutter, GitHub and Releases, Jira, Firebase App Distribution. Cloud Functions needs the Firebase Blaze plan (free quota); Donia confirms it before W3. LLM budget: see the recommendation below. Test devices: two real Android phones and one emulator.

LLM provider recommendation

Recommendation. Use **Google Gemini Flash-Lite** (currently Gemini 3.1 Flash-Lite, the cheapest current model) for the matching score and reason, called only from a Cloud Function. Keep the model name in configuration so it can be swapped. Use a hosted open-source model (Llama, Qwen or gpt-oss through a provider such as Groq or OpenRouter) as the backup option, and do not self-host.

Option	Fit for DevSync	Cost	Main risk
Gemini 3.1 Flash-Lite (recommended)	Fast and cheap for short scoring with a JSON answer; same Google/Firebase ecosystem as the backend; free tier for development	Free tier for testing; paid $0.25 per 1M input and $1.50 per 1M output tokens (Standard); Batch about half	Free-tier content is used to improve Google products, so use the paid tier once real pilot users' GitHub data is sent
Gemini 3.5 Flash-Lite or 3.7 / 3.8 Flash	Better reasoning for tuning and comparison runs	Flash-Lite 3.5: $0.30 in, $2.50 out; Flash: $0.75 in, $3.75 out (to 31 Dec 2026, then double)	Higher cost and latency than the feed needs; use only if relevance on the evaluation set is clearly better
Hosted open-source (Llama, Qwen, gpt-oss via Groq or OpenRouter)	No vendor lock-in; very fast on some hosts; good as fallback and for the evaluation set	Per-token pricing and free tiers vary by host: confirm current prices before use	Rate limits, quality differences between models and hosts, and one more API key to protect
Self-hosted open-source	Full control over data	GPU server cost, not covered by the free Firebase quota	Setup time and cost are too high for the 30 Nov deadline; not recommended

Why this fits. The matching task is short: a profile and a few projects in, a score and one-sentence reason out. A small model is enough, and the evaluation set (target +15% relevance, 50% of top-5 rated relevant) decides whether a larger model is needed. Planning estimate: one feed run of about 2,500 input and 600 output tokens costs roughly $0.0015 on Gemini 3.1 Flash-Lite, so about 6,000 runs during the pilot cost around $9. Caching results per user and limiting calls per user keep it lower.

- Call the model only from a Cloud Function; the API key stays a function secret (or use a service account if Gemini is called through Google Cloud), never in the app or repository.
- Send only public GitHub data and declared skills (NFR-12); use the paid tier for pilot users.
- Run the evaluation set on Gemini 3.1 Flash-Lite and one open-source model in W3-W4 and keep the better one; the fallback ranking from skills stays in place whichever model is chosen.
- Owner: Emad (decide by end of W1 together with Donia for the Blaze plan and budget). Prices and model names change often, so check Google's pricing page before the final budget (last checked 9 Oct 2026).

Management approach and what changed

Method. Agile Scrum with six 1-week sprints and a 4-day final window; planning every Monday with a capacity check, a 10-minute daily stand-up, a weekly review and retrospective, and a Friday supervisor update. Jira board: To do, In progress, In review, Done, linked to GitHub. Every feature is a Jira epic with sub-tasks for UI, logic, backend and tests, all owned by the same person.

Git rules. Branches: main, develop, feature/, fix/, release/. Every change is a pull request with one reviewer from the feature paper; no direct pushes to main; commits and branches start with the Jira key (for example feature/DS-12-match-feed).

Definition of done. UI uses the design system; logic and state work; Firestore data and rules in place; tests written and passing; animation on key screens (with Bishay); reviewed by a different person; tested by someone other than the author; docs written in the same pull request.

Topic	First version	Now (Feature Papers)
Ownership	Slices rotate every sprint	One owner per feature for the whole project, all layers
Workload	Not sized	S = 1, M = 3, L = 5 points; core total 47
Cover for absence	Pair work	Backup chain with a named person per owner
Authentication	Yousef	Donia (UI and logic)
Requests and FCM	Donia	Yousef
Security rules	Saga	Donia (Saga owns the report and checklist)
CI and tests setup	Saga	Yousef
Projects	No separate owner	Bishay
Design system and wireframes	Bishay	Donia did it (done); Bishay reviews and keeps brand and animations
Animations	Bishay on own screens	Bishay applies them to screens others build, through pull requests the owner reviews
Task board	Four people split by layer	Donia owns the board (5 pts); Yousef alerts, Emad dashboard, Bishay animations, Saga regression, each if time

Part 3: Working model, one owner per feature

Each person owns a feature set from screen to Firestore and tests, so everyone works on every layer. Features are sized S = 1, M = 3, L = 5 points. Core features add up to 47 points. Donia did the design system and wireframes (3 points, done), so she carries 12 including that finished work, plus project management; her 9 open points are the same as before. Bishay carries 7, Emad 8 because AI work is the hardest per point, Yousef and Saga 10 each.

Person	Hat	Core features	Core pts	If time (W5-W6)	Backup
Donia Alhosin	Project manager, data model owner	Foundation 3, Design system + wireframes 3 (done), Authentication 3, Security rules 3	12 + PM	Task board 5	Yousef
Bishay Talaat	Design and brand lead	Projects 3, Animations 3, Presentation 1	7	Extra animations + usability walkthrough 1	Emad
Yousef Mohy	App architecture lead	App shell 3, CI 1, Requests + FCM 5, Like/skip 1	10	Task alerts 3	Saga
Emad Hany	AI and prompt lead	AI matching + feed 5, Prompt evaluation 3	8	Prompt dashboard 3	Donia
Saga Maher	Quality, security and release lead	Profile + GitHub data 5, Builds + release 3, Test report + README 2	10	Regression + Pro extras 3	Bishay

Backup chain. Donia is backed up by Yousef, Yousef by Saga, Saga by Bishay, Bishay by Emad, Emad by Donia. The backup knows the feature well enough to take over.

- One owner per feature, all layers: UI, logic, Firestore, tests and docs. A different person reviews the pull request.
- Mock data first: agree the data shape in the pull request, build against sample JSON, then swap in real data.
- Monday capacity check: count each person's open points. If someone is behind, cut if-time items first, then ask the backup.

- If-time items start only after the 12 Nov core freeze and go to whoever finishes their core first. They are buffer, not a promise.
- Bishay changes other people's screens only through a pull request the screen's owner reviews.

Task assignment by feature

Feature	Owner	Size	Reviewer	Due
Foundation: repo, Firebase, data model	Donia	M, 3	Emad	22 Oct
Design system and wireframes	Donia	M, 3	Bishay	Done
App shell: Flutter, flavors, navigation	Yousef	M, 3	Bishay	22 Oct
CI and tests setup	Yousef	S, 1	Emad	22 Oct
Authentication (Google, email)	Donia	M, 3	Saga	29 Oct
Projects: seed data, add/edit	Bishay	M, 3	Emad	29 Oct
Profile + GitHub data	Saga	L, 5	Bishay	29 Oct
AI matching + match feed	Emad	L, 5	Saga	v1 29 Oct, tuned 12 Nov
Like/skip feedback	Yousef	S, 1	Emad	29 Oct
Collaboration requests + FCM	Yousef	L, 5	Saga	5 Nov
Animations on key screens	Bishay	M, 3	Donia	5 Nov
Prompt evaluation + tuning	Emad	M, 3	Yousef	12 Nov
Security rules + permissions	Donia	M, 3	Yousef	12 Nov
Obfuscated builds + release	Saga	M, 3	Donia	First build 12 Nov, final 30 Nov
Test and security report + README	Saga	S+S, 2	Bishay	12 Nov, final 30 Nov
Final presentation and demo	Bishay	S, 1	Yousef	30 Nov
Task board (if time)	Donia	5	Rotates	26 Nov
Task alerts (if time)	Yousef	3	Saga	26 Nov
Prompt dashboard (if time)	Emad	3	Donia	26 Nov
Extra animations + usability (if time)	Bishay	1	Donia	26 Nov
Regression tests + Pro extras (if time)	Saga	3	Bishay	26 Nov

The core rows total 47 points. The task board reviewer rotates, as in the first plan. The feature papers name no reviewers for the other if-time items, so those are proposed and should be confirmed at sprint planning.

Per-person timeline

Person	W1	W2	W3-W4	W5-W6 (if time)	Final
Donia	Foundation, design system and wireframes (done), Jira Sprint 1	Authentication	Security rules; PM check at 12 Nov freeze	Task board	Document features, pilot and KPI figures, final supervisor update
Bishay	Review Donia's design system; prepare Projects with mock data	Projects	Animations, due 5 Nov	Extra animations, usability walkthrough	Presentation and demo; document features
Yousef	App shell, CI	Like/skip; start request Cloud Function	Requests + FCM, due 5 Nov; help with first build	Task alerts	Document features; support Saga on release builds
Emad	AI skeleton: prompt, Cloud Function, sample JSON, evaluation set start	AI v1 and feed screens	Prompt evaluation and tuning, due 12 Nov	Prompt dashboard; tuning on real feedback; support the pilot	AI docs and final evaluation results
Saga	GitHub fetch and sample JSON; start test and security checklist	Profile screens and Firestore	Obfuscated build tested by 12 Nov; first report	Regression tests, Pro-only extras	Final builds, install link, final report, README

Handoffs: who finishes before who starts

Task	Who waits on it	Done means
Foundation: collections and field names (Donia)	Everyone	Merged to develop and written in the README
Design system and wireframes (Donia, done)	Every screen	Theme and shared widgets merged
App shell (Yousef)	All screens	App runs with stubbed routes before 22 Oct
GitHub data (Saga)	Emad's matching, profile	Fixed sample JSON returned before the real API works
Authentication user document (Donia)	Requests + FCM (Yousef)	User document with uid and FCM token field, early in W2
Seed projects (Bishay)	Emad's feed	Seed projects in Firestore
AI matching v1 (Emad)	Feed screens, prompt evaluation	Input and output JSON shape frozen
Feed card widget (Emad)	Like/skip (Yousef), same week	Card merged mid-week
Like/skip storage (Yousef)	Prompt evaluation (Emad)	Feedback documents stored in Firestore
Request document shape and send function (Yousef)	Security rules (Donia), request UI, task alerts	Shape agreed; start in W2 with a fake user document
Feed, requests, profile screens (Emad, Yousef, Saga)	Animations (Bishay)	Merged before animation pull requests
Flavors (Yousef, 22 Oct)	Release builds (Saga)	Both flavors install side by side
Task model and rules (Donia)	Task alerts (Yousef)	Collection shape and rules merged on day 1 of W5

Part 4: Risks, quality and security

Risk	Likelihood	Impact	Mitigation	Owner
Fixed deadline (30 Nov)	Medium	High	Must-have list, core freeze 12 Nov, if-time items only after it, Monday capacity check	Donia
Poor match quality	Medium	High	Evaluation set started in W1; like/skip feedback loop; fallback built first	Emad
GitHub API rate limits	Medium	Medium	Cache from the first version; batch refresh	Saga
AI cost or latency	Medium	Medium	Cache per user; limit calls per user; fallback ranking	Emad
LLM API key exposed	Low	High	Key stays a Cloud Function secret; secrets scan of the repo	Donia (scan), Emad (function)
Owner unavailable	Medium	Medium	Backup chain: Donia-Yousef, Yousef-Saga, Saga-Bishay, Bishay-Emad, Emad-Donia	Donia
Uneven load or someone falls behind	Medium	Medium	Monday capacity check; Bishay (7 pts) takes if-time work or helps Donia (12 pts) first; cut if-time items first, then ask the backup	Donia
Single-owner bottleneck on shared pieces	Medium	High	Finish data model, theme, app shell and user document in the first days; handoff table	Donia, Yousef
Animations touch others' screens	Medium	Low	Pull requests reviewed by the screen's owner	Bishay
Scope creep	Medium	Medium	If-time items start only after the 12 Nov freeze	Donia
Obfuscation breaks the release build	Low	Medium	Test an obfuscated build by 12 Nov	Saga
Install link hard to use	Low	Medium	5-line install guide; test on a clean device	Saga
Merge conflicts	High	Low	Small branches, frequent pull requests, CI blocks failing tests	Yousef
LLM price, limits or terms change	Medium	Medium	Model name kept in config; evaluation set run on Gemini Flash-Lite and one open-source model; fallback ranking	Emad
Firebase misconfiguration or cost	Low	High	Rules review; dev and prod projects; budget alerts; confirm Blaze plan before W3	Donia

Testing. Unit tests for matching logic and data models; widget tests for key screens; integration tests for sign-in, feed and requests; prompt evaluation set; two real Android devices and one emulator; every feature tested by someone other than

its author before merge. Saga owns the test and security report and regression runs.

Security. Runtime permissions only when needed and explained; Firebase UI Auth; Firestore rules so users read and write only their own data and their requests (Donia); no secrets in the repo; HTTPS only; obfuscated release builds with symbol files stored safely (Saga); security checklist completed before release.

Part 5: KPIs and course requirements

Targets are proposed and can be adjusted with the supervisor. KPI owners now follow feature ownership.

KPI	Target	Measured by	Owner
Match response time	5 s or less for 95% of requests	Timing logs, Performance Monitoring	Emad
App start time	Cold start 3 s or less on test devices	Manual timing, Performance Monitoring	Yousef
Notification delivery	FCM alert within 10 s in 95% of tests	Delivery test script with timestamps	Yousef
System uptime	99.5% or higher during testing and pilot	Firebase status and monitoring	Donia
Stability	99% or more crash-free sessions	Firebase Crashlytics	Saga
Test quality	All tests pass on main; 70% or more coverage of matching logic; 0 critical bugs at release	CI results, Jira bug list	Saga
Security	100% of the security checklist	Checklist review before release	Saga (rules items by Donia)
Match quality	At least 50% of top-5 matches rated relevant	Like/skip feedback	Emad
Prompt improvement	Relevance up 15% or more, first to final version	Evaluation set results	Emad
User adoption	At least 20 pilot users; 60% complete their profile	Firebase Auth and Firestore counts	Donia
Collaboration activity	At least 10 requests sent during the pilot	Firestore request records	Donia
Sprint delivery	80% or more of sprint tasks completed each sprint	Jira sprint reports	Donia
Milestones	All milestones on time; 100% of must-have scope	Milestone review, Jira	Donia
Load balance (replaces team coverage)	Core points: Donia 12 (3 done), Bishay 7, Yousef 10, Emad 8, Saga 10; every owner has UI, logic, Firestore and tests done for their features by 12 Nov	Jira epics and points	Donia

Course requirements checklist

Requirement	Lead	Due
Git and GitHub with branching and pull requests	Donia	From 16 Oct
Firebase UI Auth (Google, email)	Donia	29 Oct
Free and Pro flavors	Yousef	Flavors 22 Oct, builds 30 Nov
FCM notifications	Yousef	5 Nov
Implicit and advanced animations	Bishay	5 Nov
Security best practices and runtime permissions	Donia (rules, permissions), Saga (checklist)	12 Nov
Code obfuscation and Android App Bundle	Saga (Yousef supports)	First build 12 Nov, final 30 Nov
Technical documentation and README	Saga assembles; each owner writes their own	30 Nov

Part 6: Future work

These items are outside the 30 Nov scope.

Phase	Idea	Why it helps
Next (Dec 2026)	Pilot with 20+ users and collect like/skip data	Real evidence for match quality and adoption KPIs
Next	Automated prompt evaluation in CI; Remote Config prompt versions with A/B tests	Every prompt change is measured; switch versions without a release
Next	Firebase App Check and basic rate limiting	Protects backend and LLM budget
Next	Grow the task board into a full task management system	Keeps matching, requests and delivery in one place

Phase	Idea	Why it helps
Later	Web platform, iOS build, real Pro plan with payments	Wider reach and a business model
Later	Issue-level matching, skill-gap suggestions; maintainer search, in-app chat, endorsements	Moves from project to task; completes the two-sided marketplace